

Chapter 31-Spanning Tree Protocol (STP) – Loop Prevention & Redundancy

1 Introduction to STP

What is STP?

Spanning Tree Protocol (STP) is a Layer 2 protocol used to:

- ✓ Prevent network loops
 - ✓ Avoid broadcast storms
 - ✓ Maintain redundancy in networks
 - 📌 Standard: **IEEE 802.1D**
-

2 Redundancy in Networks

Why Redundancy is Important?

Modern networks must run:

🕒 **24/7/365** (always ON)

Even small downtime = ❌ business loss

Problem: Poor Network Design

If network has:

- ❌ Single connection
- ❌ Single switch
- ❌ Single path

👉 Then **failure = network down**

This is called:

● **Single Point of Failure**

✓ Better Network Design

- ✓ Multiple switches
- ✓ Multiple links
- ✓ Backup paths

This is called:

● Redundancy

Real Life Example

Think of **roads in a city** 🚗

- One road → traffic jam if blocked ❌
- Multiple roads → traffic reroutes ✓

Same in networking:

- Multiple paths = **backup available**
-

⚠️ 3 Problem with Redundancy

Redundancy creates **loops** 🤪

🔥 Broadcast Storm

Ethernet frames:

- ❌ No TTL field
- 👉 Packets never expire

So broadcast traffic:

- 🔄 Loops forever
- 📈 Increases traffic
- 💥 Network crash

Example

Scenario:

- 3 switches connected
- PC1 sends data to PC2

Flow:

- 1 PC1 → SW1
- 2 SW1 → SW2 & SW3
- 3 SW3 again sends back to SW1

👉 Infinite loop 

MAC Address Flapping

Switch learns MAC address like:

MAC → Port mapping

But in loop:

- Same MAC seen on different ports
- Switch keeps updating table

👉 This is called:

⚠️ **MAC Address Flapping**



Solution: STP (Spanning Tree Protocol)

STP solves loops by:

👉 Blocking some ports



How STP Works

- ✓ Creates **loop-free topology**
 - ✓ Selects only ONE path
 - ✓ Blocks redundant paths
-



Port States in STP

State	Description
Forwarding	Active traffic
Blocking	Disabled (backup)



Important Rule

Blocked ports:

- ✗ Do NOT forward data
 - ✓ Only send/receive STP messages (BPDUs)
-



5 BPDU (Bridge Protocol Data Unit)

STP uses **BPDU** messages.



Sent every **2 seconds**



Purpose of BPDU

- ✓ Exchange network info
 - ✓ Elect Root Bridge
 - ✓ Maintain topology
-



Important

Only **switches** send BPDUs



PCs and routers DO NOT



6 Bridge = Switch

In STP:



Switch is called **Bridge**



7 Root Bridge Concept



What is Root Bridge?



Main switch in network

All traffic decisions are based on it.



Root Bridge Election

Switch with:



Lowest Bridge ID wins

Bridge ID =

Bridge Priority + MAC Address

Default Values

- Priority = **32768**
 - So lowest MAC wins by default
-

Important Rule

- ✓ Compare Priority first
 - ✓ If same → compare MAC address
-

Extended System ID

Bridge Priority + VLAN ID

Example:

VLAN 1:

$32768 + 1 = 32769$

Bridge Priority Values

Can change only in steps of **4096**

Valid Values:

0, 4096, 8192, 12288, 16384, ...

PVST (Cisco STP)

Cisco uses:

- ◆ **PVST (Per VLAN Spanning Tree)**

- 👉 Each VLAN runs its own STP

- ✓ Different root bridge per VLAN possible

1 STP Working Steps

Step 1: Root Bridge Election

- ✓ Lowest Bridge ID wins

- ✓ All ports = **Designated Ports (Forwarding)**

Step 2: Root Port Selection

Each switch selects:

- 👉 ONE port → Root Port

Selection Rules:

-  Lowest Root Cost
 -  Lowest Neighbor Bridge ID
 -  Lowest Port ID
-

Step 3: Designated Port Selection

Each link chooses:

- 👉 ONE Designated Port

Other becomes:

- ✗ Non-Designated (Blocking)

1 STP Cost Chart

Speed Cost

10 Mbps 100

100 Mbps 19

1 Gbps 4

10 Gbps 2



Important Rule

Cost calculated:

- Only on **outgoing interfaces**
- Add cost till reaching root bridge



1 Port Roles

Role

Function

Root Port Best path to root

Designated Port Forwarding on segment

Non-Designated Blocking



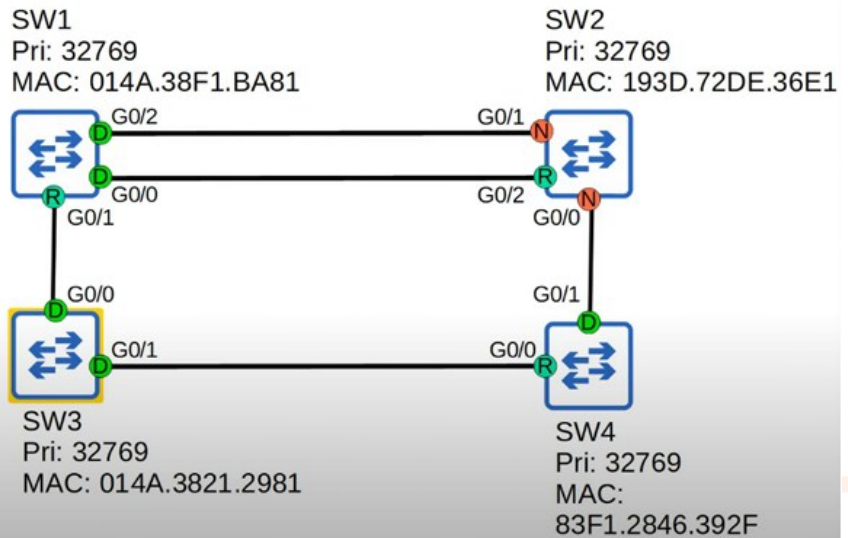
1 Port ID

Port ID = Priority (128 default) + Port Number

Used for tie-breaking.

1 QUIZ 1 – Topology Explanation

Identify the root bridge, and the role of each interface on each switch in the network (root/designated/non-designated)



Given:

All switches have same priority → 32769

So decision based on:

👉 Lowest MAC Address

🏆 Root Bridge

✓ **SW3** (lowest MAC)

🔌 Port Roles

SW3 (Root Bridge)

✓ All ports → Designated (Forwarding)

SW1 & SW4

- Ports connecting to SW3 → Root Ports

SW2

- Chooses best path to root → Root Port
- Remaining ports → Blocking

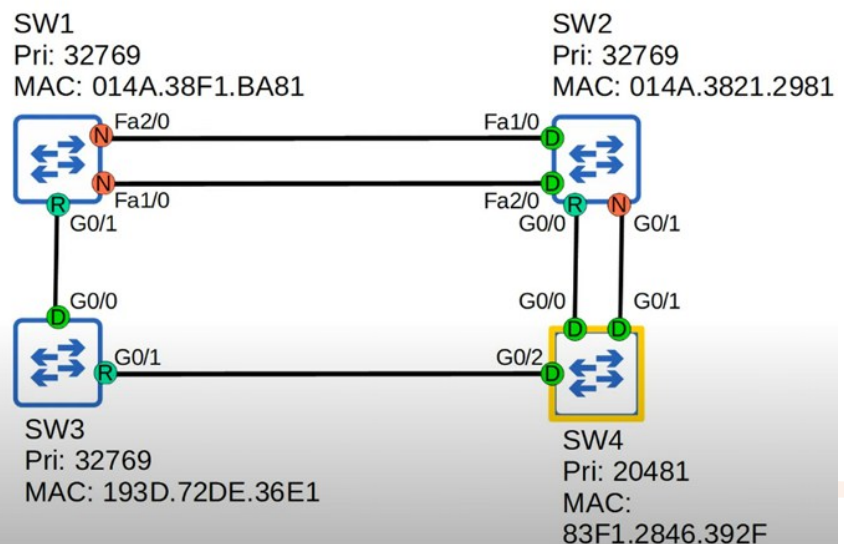
⚠ Important Concept

If two paths exist:

👉 Lower port number wins

1 5 QUIZ 2 – Topology Explanation

Identify the root bridge, and the role of each interface on each switch in the network (root/designated/non-designated)



🔍 Given:

SW4 priority = **20481 (lowest)**

🏆 Root Bridge

✓ SW4

Port Roles

SW4

✓ All ports → Designated

SW2

✓ G0/0 → Root Port (lower port number)

SW3

✓ G0/1 → Root Port

SW1

✓ G0/1 → Root Port

✗ Fa ports → Blocking (higher cost)

Rule Used

✓ Lowest cost wins

✓ If tie → port number

1 STP States

BLOCKING (Stable)

✗ No traffic

✗ No MAC learning

✓ Only BPDU

2 LISTENING (15 sec)

- ✓ Only BPDU
 - ✗ No traffic
 - ✗ No MAC learning
-

3 LEARNING (15 sec)

- ✓ Learns MAC
 - ✗ No traffic
-

4 FORWARDING (Stable)

- ✓ Full traffic
 - ✓ MAC learning
 - ✓ BPDU
-



Timers

Timer	Value
Hello	2 sec
Forward Delay	15 sec
Max Age	20 sec



Final Understanding

✓ What STP Does

- ✓ Prevents loops
- ✓ Maintains redundancy
- ✓ Blocks extra paths
- ✓ Activates backup when needed

Real Life Example

Think of **railway tracks** 🚂

- Many tracks exist (redundancy)
- Only one active track at a time
- Others are backup

STP works the same way.

Final Summary

Topic	Key Point
STP	Prevents Layer 2 loops
Root Bridge	Main switch
BPDU	Control message
Port Roles	Root, Designated, Blocking
States	Blocking → Listening → Learning → Forwarding
Cost	Path selection



1 Chapter Summary (Quick Revision)



What is STP?

Spanning Tree Protocol (STP) is a Layer 2 protocol used to:

- ✓ Prevent loops
- ✓ Avoid broadcast storms
- ✓ Maintain redundancy



Standard: **IEEE 802.1D**



Why Redundancy is Needed?

Modern networks must run:



24/7/365

To avoid:



Single Point of Failure



Without Redundancy

One link fails → Entire network down



With Redundancy

Multiple links → Backup available



Problem: Loops in Network

When we add redundancy:



Loops are created



Broadcast Storm

- Ethernet has **no TTL**

- Broadcast frames loop forever

Result:

-  Traffic increases
-  Network crash

MAC Address Flapping

- Same MAC seen on multiple ports
- Switch keeps updating MAC table

 This is called **MAC Flapping**

Solution: STP

STP creates a **loop-free topology**

- ✓ Blocks redundant paths
- ✓ Keeps only one active path

STP Port States (Simplified)

State	Description
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Forwarding	Active traffic
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Blocking	Backup path
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BPDU (Important Concept)

- STP uses **BPDU (Bridge Protocol Data Unit)**
 - Sent every **2 seconds**
- ✓ Elect root bridge
 - ✓ Maintain topology
-  Only switches send BPDUs

Root Bridge (Most Important Concept)

👉 Main switch in network

Election Rule:

Lowest **Bridge ID** wins

Bridge ID

Bridge ID = Priority + MAC Address

Default:

- Priority = 32768
 - Lowest MAC wins
-

Bridge Priority

Valid values:

0, 4096, 8192, 12288, ...

PVST (Cisco)

👉 Per VLAN STP

✓ Each VLAN has its own STP

✓ Different root per VLAN

STP Working Steps

Step 1: Root Bridge Election

Lowest Bridge ID wins

Step 2: Root Port Selection

Each switch selects:

 One **Root Port**

Rules:

- 1** Lowest cost
- 2** Lowest neighbor bridge ID
- 3** Lowest port ID

Step 3: Designated Port Selection

Each link:

✓ One **Designated Port (Forwarding)**

✗ Others = Blocking

STP Cost Table

Speed	Cost
-------	------

10 Mbps	100
---------	-----

100 Mbps	19
----------	----

1 Gbps	4
--------	---

10 Gbps	2
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Port Roles

Role	Meaning
Root Port	Best path to root
Designated Port	Forwarding port
Non-Designated	Blocking

STP States (Detailed)

State	Time	Description
Blocking	Stable	No traffic
Listening	15 sec	BPDU only
Learning	15 sec	MAC learning
Forwarding	Stable	Full traffic

Timers

Timer	Value
Hello	2 sec
Forward Delay	15 sec
Max Age	20 sec

Chapter Conclusion

STP is **critical for Layer 2 network stability**.

Without STP:

- ✗ Loops
- ✗ Broadcast storms
- ✗ Network crash

With STP:

- ✓ Loop-free topology
- ✓ Redundancy maintained
- ✓ Backup links available

 STP intelligently blocks and activates paths to ensure **safe and efficient communication**.

Q & A

1 What is STP?

Answer

Spanning Tree Protocol (STP) is a Layer 2 protocol used to prevent loops in a switched network.

It creates a loop-free topology by blocking redundant paths while still maintaining redundancy.

2 Why is STP required?

Answer

Because redundancy creates loops, which leads to:

-  Broadcast storms
-  MAC flapping
-  Network instability

STP prevents these issues.

3 What is a Broadcast Storm?

Answer

A broadcast storm occurs when broadcast frames loop infinitely in the network.

Reason:

-  Ethernet frames do not have TTL

Impact:

-  Network congestion
 -  Switch CPU high
 -  Network crash
-

4 What is MAC Address Flapping?

Answer

When a switch sees the same MAC address on multiple ports repeatedly, it keeps updating the MAC table.

This is called MAC flapping and indicates a loop in the network.

5 What is BPDU?

Answer

BPDU (Bridge Protocol Data Unit) is a control message used by STP.

Functions:

- ✓ Root bridge election
- ✓ Topology maintenance

Sent every **2 seconds**

6 What is Root Bridge?

Answer

Root Bridge is the central switch in STP topology.

All path calculations are based on it.

7 How is Root Bridge elected?

Answer

Based on **lowest Bridge ID**:

- 1 Lowest Priority
 - 2 If tie → Lowest MAC address
-

8 What is Bridge ID?

Answer

Bridge ID = Priority + MAC Address

Used to elect root bridge.

9 What is Root Port?

Answer

Root Port is the port on a switch that has the best path to the root bridge.

Each non-root switch has only one root port.

10 What is Designated Port?

Answer

Designated port is the port that forwards traffic on a network segment.

Each segment has one designated port.

1 1 What is Blocking Port?

Answer

Blocking port does not forward traffic.

It is used to prevent loops and acts as a backup.

1 2 Explain STP states

Answer

- 1** Blocking – No traffic
 - 2** Listening – BPDU exchange
 - 3** Learning – MAC learning
 - 4** Forwarding – Normal traffic
-

1 3 What is STP cost?

Answer

Cost determines best path to root bridge.

Lower cost = better path

1 4 What is PVST?

Answer

Per VLAN Spanning Tree (PVST) is Cisco's implementation of STP.

Each VLAN runs its own STP instance.

1 5 What happens if STP is disabled?

Answer

Network will suffer from:

- ✳ Broadcast storms
 - ✳ MAC flapping
 - ✳ Network crash
-

1 6 How to check STP status?

show spanning-tree

1 7 How to change root bridge?

spanning-tree vlan 1 priority 4096

Lower priority = higher chance to become root

1 8 How to troubleshoot STP?

Steps:

- ✓ Check root bridge
 - ✓ Verify port roles
 - ✓ Check blocked ports
 - ✓ Analyze cost
-

 SCENARIO QUESTIONS

1 9 Scenario

Network is slow and CPU is high

Answer:

Possible reason:

👉 Broadcast storm due to loop

Check:

- STP status
 - Redundant links
-

2 0 Scenario

MAC address keeps changing

Answer:

👉 MAC flapping

Reason:

Loop in network

2 1 Scenario

Two paths available – which will STP choose?

Answer:

STP chooses path based on:

- 1** Lowest cost
 - 2** Lowest bridge ID
 - 3** Lowest port ID
-

2 2 Scenario

SVI working but network unstable

Answer:

Check STP blocking ports
Possible loop issue

2 3 What is Extended System ID?

Answer

Bridge Priority + VLAN ID

Example:

32768 + VLAN 1 = 32769

2 4 Why STP convergence is slow?

Answer

Because of:

-  Listening + Learning (30 sec total)
-

2 5 What is convergence?

Answer

Time taken by network to reach stable topology after change.

 One Liner

 “Explain STP in simple words”

Say:

“STP is a Layer 2 protocol that prevents loops in a network by blocking redundant paths while keeping backup links ready.”

 FINAL REVISION POINTS

- ✓ STP prevents loops
 - ✓ Root bridge is central device
 - ✓ BPDU used for communication
 - ✓ Ports: Root / Designated / Blocking
 - ✓ States: Blocking → Listening → Learning → Forwarding
 - ✓ Cost decides best path
-